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Section: BSIT – 4C

Enterprise Infrastructure Plan

PART 1 — Company Profile

1. Company Name

- ABC Startup Solutions

2. Nature of Business

- ABC Startup Solutions is a small software development company that creates websites, software applications, and other IT solutions for businesses. The company also provides technical support and basic IT services to its clients.

3. Company Vision

- To become a trusted software company that provides useful and reliable technology solutions.

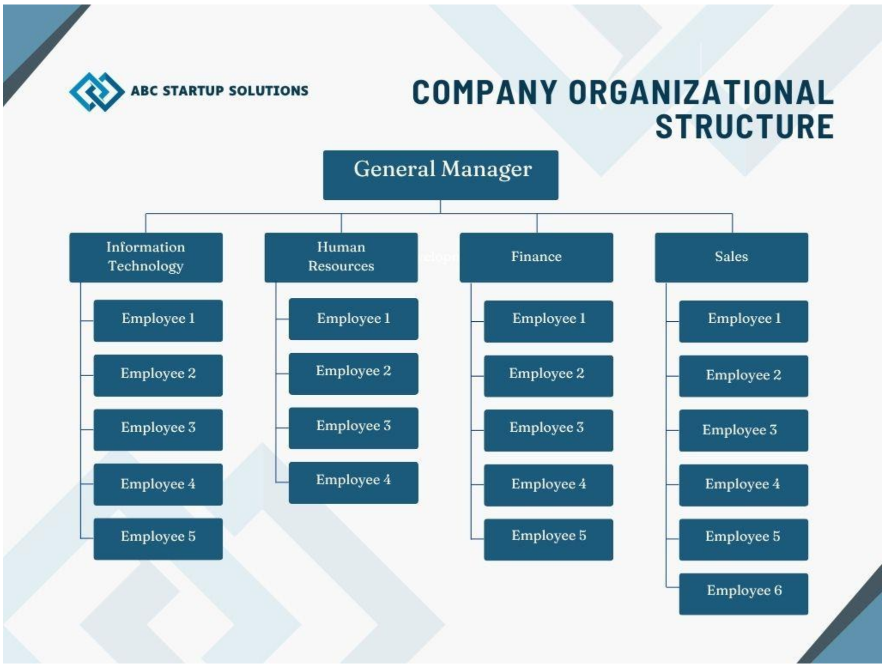
4. Company Mission

- To provide quality and affordable software solutions while giving good service to our clients and creating a good working environment for our employees.

5. Office Location

- Unit 113, ABC Business Center, National Highway, Santa Rosa, Laguna, Philippines

6. Organizational Structure



7. Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	**20**

PART 2 — Enterprise Hardware Inventory

The following hardware inventory identifies the equipment required to support the daily operations of ABC Startup Solutions. The inventory covers employee workstations, servers, networking equipment, backup devices, and other essential IT hardware.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computers	15	IT, HR, Finance, Sales	Main computers for employees who work mostly in the office
HW-002	Laptops	5	IT, Sales	For IT staff and employees who may need to work outside their desks or attend meetings
HW-003	Server	1	IT	Used for centralized files, user accounts, applications, and other company services
HW-004	Network Switch	2	IT	Connects computers, server, printer, and other network devices
HW-005	Router	1	IT	Provides internet access and manages the company's network
HW-006	Printer	2	HR, Finance	Used for printing documents, reports, forms, and other office files
HW-007	UPS	4	IT, Finance, Server Room	Provides temporary power during power interruptions and protects equipment
HW-008	Wireless Access Point	3	All Departments	Provides Wi-Fi connection throughout the office
HW-009	NAS Storage	1	IT	Used for shared file storage and an additional backup location

HW-010	External Backup Drive	2	IT	Used for offline backups of important company files and server data
HW-011	Monitors	20	All Departments	Provides a display for each employee's workstation

Quantity Justification

- **15 Desktop Computers** – Most employees will work in the office, so desktops will be used as their main workstations.
- **5 Laptops** – These can be assigned mainly to IT employees and some Sales employees who may need to work during meetings or outside the office.
- **1 Server** – Since the company only has 20 employees, one server is enough for basic company services and centralized storage.
- **2 Network Switches** – Two switches provide enough ports for the computers, server, printer, access points, and other network devices, with extra ports for future expansion.
- **1 Router** – One business-grade router is enough to manage the company's internet connection and local network.
- **2 Printers** – One can be assigned near HR and another near Finance since these departments are more likely to print documents.
- **4 UPS Units** – UPS units will protect the server, network equipment, and important workstations from sudden power interruptions.
- **3 Wireless Access Points** – Three access points can provide Wi-Fi coverage across the single office floor and help avoid weak signals.
- **1 NAS Storage** – A NAS provides centralized storage for shared company files and can also be used for backups.
- **2 External Backup Drives** – Having two drives allows the company to keep backup copies separately and rotate them for better protection.
- **20 Monitors** – Each of the 20 employees will have a monitor for their workstation. Laptop users can also connect their laptops to these monitors when working in the office.

PART 3 — Enterprise Software Inventory

The following software inventory identifies the applications and operating systems required to support the daily operations of ABC Startup Solutions. The selected software supports productivity, software development, server management, security, file management, and technical support.

Software	Version	License	Purpose
Windows 11 Pro	Windows 11 Pro 24H2	Paid	Main operating system for employee computers

Ubuntu Server	Ubuntu Server 24.04 LTS	Free / Open Source	Used for the company's server and network services
Microsoft Office	Microsoft 365 Apps	Subscription	Used for documents, spreadsheets, presentations, and other office tasks
Visual Studio Code	Latest Stable	Free	Used by IT employees for coding and software development
Git	Latest Stable	Free / Open Source	Used to track and manage changes in source code
GitHub Desktop	Latest Version	Free	Makes it easier to manage GitHub repositories and upload code
VirtualBox	Latest Version	Free / Open Source	Used to create virtual machines for testing different operating systems
Google Chrome	Latest Stable	Free	Main web browser for internet access and web-based applications
Microsoft Defender	Built-in / Latest	Included with Windows	Protects computers from viruses, malware, and other security threats
AnyDesk	Latest Version	Free / Paid License	Used by IT staff for remote technical support
7-Zip	Latest Stable	Free / Open Source	Used to compress and extract files and folders

Why Each Software Is Needed

- **Windows 11 Pro** – Used as the main operating system for the company's desktop computers and laptops. The Pro version also provides additional business and security features.
- **Ubuntu Server** – Used for the company server because it is free, reliable, and suitable for hosting different network and server services.
- **Microsoft Office** – Needed by HR, Finance, Sales, and IT for creating documents, reports, spreadsheets, presentations, and other office files.
- **Visual Studio Code** – Important for the IT department because it provides tools for writing and editing programming code.
- **Git** – Helps developers keep track of changes in their projects and allows them to manage different versions of their code.
- **GitHub Desktop** – Provides an easier way for employees to work with GitHub repositories without relying only on command-line commands.
- **VirtualBox** – Allows the IT department to create virtual computers for testing software, operating systems, and server configurations without affecting the main systems.
- **Google Chrome** – Used for browsing the internet, accessing online services, testing websites, and using web-based applications.
- **Microsoft Defender** – Provides basic antivirus and security protection against malware and other common threats.

- **AnyDesk** – Allows IT staff to remotely access computers when troubleshooting or providing technical support.
- **7-Zip** – Used to compress large files and extract downloaded or archived files, which can also help save storage space.

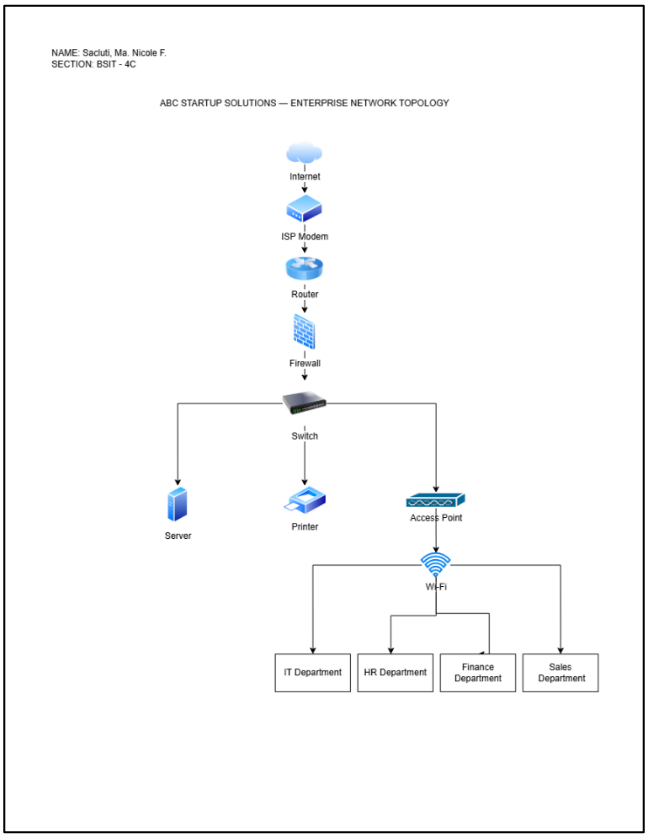
PART 4 — Enterprise Network Inventory

The following network equipment will be used to connect the 20 employees of ABC Startup Solutions. The setup provides internet access, wired and wireless connectivity, and basic network security.

Asset ID	Network Equipment	Quantity	Location
NET-001	ISP Modem	1	Server Room
NET-002	Router	1	Server Room
NET-003	Firewall	1	Server Room
NET-004	Network Switch	2	Server Room
NET-005	Wireless Access Point	3	Office Area
NET-006	Patch Panel	1	Server Room
NET-007	CAT6 Ethernet Cable	500 meters	Office / Server Room
NET-008	RJ45 Connectors	100 pieces	IT Storage

PART 5 — Enterprise Network Diagram

The network diagram shows how the company's devices and departments are connected to provide reliable internet access and communication.



PART 6 — System Administration Roles

1. **Helpdesk Technician**

- A Helpdesk Technician provides technical assistance to employees who experience problems with their computers, software, internet connection, printers, or other IT equipment. They are usually the first person users contact when they have a technical problem.

Responsibilities

- Troubleshoot computer hardware and software problems
- Assist employees with technical issues
- Install and configure software and devices
- Set up new computers and user accounts
- Help users with printers, email, and other applications
- Document and track reported problems
- Escalate difficult problems to higher-level IT staff

Skills

- Basic computer hardware and software knowledge
- Troubleshooting and problem-solving
- Good communication skills
- Customer service skills
- Basic networking knowledge
- Patience and ability to explain technical issues clearly

Common Tools

- Windows 11
- Microsoft 365
- Google Chrome
- AnyDesk
- Antivirus software
- Helpdesk/ticketing systems
- Remote support tools
- Command Prompt

Certifications

- **CompTIA A+** – covers basic IT support and troubleshooting skills.
- **CompTIA Network+** – provides basic networking knowledge.
- **ITIL Foundation** – introduces IT service management concepts.

2. **Network Administrator**

- A Network Administrator is responsible for setting up, maintaining, monitoring, and troubleshooting an organization's computer network. This includes equipment such as routers, switches, servers, and other network devices.

Responsibilities

- Install and configure network equipment
- Maintain LAN and internet connections

- Monitor network performance
- Troubleshoot network problems
- Manage routers, switches, and access points
- Maintain network security
- Manage user access and permissions
- Perform network upgrades and maintenance
- Help ensure that the network is available and working properly

Skills
- Networking fundamentals
- IP addressing and subnetting
- Router and switch configuration
- Network troubleshooting
- Network security
- Knowledge of LAN and WAN
- Problem-solving skills
- Basic server administration

Common Tools
- Cisco Packet Tracer
- Wireshark
- Ping
- Traceroute
- Command Prompt
- PowerShell
- Network monitoring tools
- Routers and switches
- Firewall management tools

Certifications
- **Cisco Certified Network Associate (CCNA)** – covers networking fundamentals, network access, IP connectivity, IP services, security fundamentals, and automation.
- **CompTIA Network+** – focuses on networking concepts, troubleshooting, security, and network infrastructure.
- **Cisco Certified Network Professional (CCNP)** – a more advanced certification for networking professionals.

3. Linux System Administrator

- A Linux System Administrator is responsible for managing and maintaining computers and servers that use Linux operating systems. They make sure that systems are working properly, secure, updated, and available for users. They also handle user accounts, software installation, system configurations, troubleshooting, and regular maintenance.

Responsibilities
- Install, configure, and maintain Linux servers
- Manage users, groups, and permissions
- Install and update software packages
- Monitor system performance and services

- Manage storage and file systems
- Troubleshoot Linux system and network problems
- Maintain system security and backups

Skills

- Knowledge of Linux operating systems
- Command-line and shell scripting skills
- Basic networking knowledge
- File and permission management
- System troubleshooting
- User and group administration
- Basic security knowledge

Common Tools

- Ubuntu Server
- Red Hat Enterprise Linux
- Bash
- SSH
- systemctl
- Cron
- top / htop
- vim / nano
- Docker

Certifications

- **Red Hat Certified System Administrator (RHCSA)** – validates core Linux system administration skills.
- **Red Hat Certified Engineer (RHCE)** – an advanced certification for Red Hat system administration and automation.
- **Linux Foundation Certified System Administrator (LFCS)** – focuses on practical Linux administration skills.

4. Cloud Administrator

- A Cloud Administrator is responsible for managing and maintaining an organization's cloud resources and services. They make sure that cloud systems are available, secure, and working properly. They also manage cloud storage, virtual machines, user access, network settings, backups, and monitor the performance of cloud services.

Responsibilities

- Manage cloud servers and virtual machines
- Configure cloud storage and networks
- Manage users, permissions, and access
- Monitor cloud resources and performance
- Apply security settings and policies
- Manage backups and recovery
- Monitor and control cloud costs
- Troubleshoot cloud-related problems

Skills
- Knowledge of cloud computing
- Networking and virtualization
- Cloud security
- Storage management
- User and access management
- Troubleshooting
- Basic scripting and automation
- Knowledge of cloud platforms such as Azure or AWS

Common Tools
- Microsoft Azure Portal
- Azure CLI
- PowerShell
- AWS Management Console
- AWS CLI
- Microsoft Entra ID
- Azure Monitor
- AWS CloudWatch

Certifications
- **Microsoft Certified: Azure Administrator Associate (AZ-104)** – focuses on managing Azure identities, storage, compute, networking, monitoring, and recovery.
- **AWS Certified Cloud Practitioner** – covers basic AWS cloud concepts and services.
- **AWS Certified Solutions Architect** – Associate – focuses on designing and implementing solutions on AWS.
- **Google Associate Cloud Engineer** – focuses on deploying and managing applications and infrastructure on Google Cloud.

The Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator work together to keep the organization's IT systems running properly. The Helpdesk Technician assists employees with their technical problems, while the Network Administrator manages the company's network and connections. The Linux System Administrator maintains Linux servers and makes sure they are secure and working properly. The Cloud Administrator manages the company's cloud services, storage, and resources. By working together, they can solve IT problems faster and keep the company's systems reliable and secure.

PART 7 — Infrastructure Recommendations

The following recommendations are proposed for ABC Startup Solutions to provide a reliable, secure, and manageable IT infrastructure. Since the company currently has 20 employees, the recommendations are focused on meeting its current needs while allowing room for future growth.

1. **Internet Provider**

- The company should use a reliable fiber internet provider with at least 300–500 Mbps speed. This should be enough for browsing, software development, video meetings, file sharing, and cloud applications. It would also be better to have a backup internet connection in case the main connection stops working.

2. **Server Specifications**

- The company should have a server with at least 32 GB RAM, 2 TB SSD storage, and a reliable server-grade processor. The server can use Ubuntu Server and can be used for storing company files, managing services, and other IT tasks. This setup should be enough for the company's current number of employees.

3. **Backup Strategy**

- The company should regularly back up important files and server data. A 3-2-1 backup strategy can be used, where there are three copies of important data, stored in two different types of storage, with one copy kept in another location. The company can use the NAS and external backup drives for this purpose. Backups should be done regularly and checked to make sure the files can still be recovered.

4. **Security Recommendations**

- The company should use a firewall, secure Wi-Fi, strong passwords, user access controls, and regular software updates. Employees should also receive basic security training so they know how to avoid phishing emails, suspicious links, and malware. Only authorized employees should be given access to important company files and systems.

5. **Antivirus**

- All company computers should have Microsoft Defender Antivirus enabled. It is already included with Windows 11 and provides basic protection against viruses and malware. The IT department should make sure that Defender is updated and that real-time protection is always turned on.

6. **Password Policy**

- Employees should use strong and unique passwords for their accounts. Passwords should not be shared with other employees or reused for different accounts. Multi-factor authentication (MFA) should also be enabled for important accounts, especially administrator, email, and cloud accounts. This will provide an additional layer of protection.

7. **Expansion Plan**

- The company should leave room for future expansion. For example, network switches should have extra ports, and the server should be upgradeable with additional RAM and storage. More computers, access points, and other equipment can be added when the company hires more employees. Cloud services can also be considered as the company grows.

PART 8 — Personal Reflection

Doing this project gave me a better understanding of what a System Administrator actually does in a company. Before starting the project, I thought that system administration was mostly about fixing computers, setting up networks, and solving technical problems. As I worked on each part, I realized that there is a lot of planning involved before any equipment or software is installed. I learned how to identify the hardware and software a company needs, plan a network, understand different IT roles, and consider security and backup solutions. I also learned that every decision should be based on the size and needs of the company.

The most challenging part for me was creating the enterprise network diagram. I had to figure out how the internet, modem, router, firewall, switch, server, printer, access points, and departments should be connected. It was also important to make the diagram organized and easy to understand. At first, I was not completely sure where some of the devices should be placed, but working through the network setup helped me understand how the different devices communicate with each other.

I also learned why planning is important before deploying an IT infrastructure. Without proper planning, a company could spend money on equipment that it does not need or forget something important. Planning also helps prevent problems with network connections, security, storage, and backups. It gives the company a clear idea of what equipment and resources are needed before the actual setup begins. It also makes it easier to prepare for future growth.

This project can help me become a better System Administrator because it allowed me to practice planning an IT infrastructure instead of only focusing on individual computers or devices. It taught me to look at the bigger picture and think about how everything works together. I also learned that a good System Administrator needs to be organized, patient, and willing to troubleshoot problems. Overall, this project helped me understand the responsibilities of the job better and gave me useful knowledge that I can apply to future projects and actual IT work.